

DEVELOPMENT PACKAGE 0

AI Development Protocol

Project: Bidra Platform

Version: 1.0

Purpose: Universal execution rules for every AI coding session

1. Objective

This document defines the mandatory engineering rules that every AI coding session must follow.

It applies to every Development Package.

No package may override these rules.

These rules exist to guarantee architectural consistency across the entire Bidra platform.

2. Before Writing Code

The AI must first:

- Read the current Development Package.
- Read the referenced Volumes.
- Inspect the existing codebase.
- Understand current architecture.
- Identify affected modules.
- Produce a short implementation plan.

Do **not** begin coding immediately.

3. Never Rewrite Working Code

Never replace existing implementations unless:

- fixing a verified bug

- improving performance
- improving security
- explicitly instructed

Prefer extending existing modules.

4. Respect Module Boundaries

A module owns its own domain.

Examples:

Authentication owns:

- Users
- Sessions
- Roles
- Permissions

Organizations own:

- Organizations
- Members
- Verification

Campaigns own:

- Campaigns
- Needs

Payments own:

- Payments
- Ledger

Volunteer owns:

- Attendance
- Wallet
- Time Credits

Never place business logic in another module.

5. Database Rules

Never:

- bypass Prisma
- write raw SQL unless necessary
- duplicate entities
- duplicate business rules

Every schema change requires:

- migration
 - Prisma update
 - tests
 - documentation
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6. API Rules

Every endpoint must include:

- validation
- authorization
- typed DTOs
- OpenAPI documentation
- standard error handling
- audit logging where required

No undocumented endpoint is allowed.

7. Frontend Rules

Every page must include:

- loading state
- empty state
- error state
- responsive layout
- accessibility
- permission awareness

Never hardcode API responses.

8. Security Rules

Never:

- trust frontend validation
- expose UUIDs publicly
- log secrets
- expose password hashes
- bypass authorization

All permissions are enforced server-side.

9. Financial Rules

Money is immutable.

Never:

- edit ledger entries
- delete payments
- mutate financial history

Corrections create new ledger entries.

10. Time Credit Rules

Time Credits are immutable.

Wallet balance is always derived from ledger entries.

Never edit wallet balances directly.

11. Testing Rules

Every feature requires:

- unit tests
- integration tests
- API tests
- frontend tests where applicable

Bug fixes require regression tests.

12. Documentation Rules

Every completed feature updates:

- OpenAPI
- database documentation
- developer documentation
- architecture diagrams if affected

Documentation must always reflect reality.

13. AI Self-Review

Before completion, verify:

- architecture respected
- no duplicated logic
- permissions enforced
- tests pass
- lint passes
- type checks pass
- documentation updated

Only then mark the task complete.

14. Git Rules

Each package should be delivered as a separate branch and Pull Request.

One package equals one merge.

Avoid mixing unrelated functionality.

15. Final Principle

When in doubt:

- prefer simplicity

- prefer readability
- prefer consistency
- prefer maintainability

Optimize for the next five years of development, not just the next feature.